PRESIDENTS OFFICE

REGIONAL ADMINSTRATION AND LOCAL GOVERNMENT

ARUSHA REGION MOCK EXAMINATION MARKING SCHEME

CHEMISTRY 1.

SECTION A (16 Marks)

1.

Question	i	ii	iii	iv	v	vi	vii	viii	ix	x
Answer	C	C	B	D	C	D	A	E	C	D

1@=10

2.

LIST A	i	ii	iii	iv	v	vi
LIST B	F	E	C	D	G	B

1@=06

SECTION B

3. (a) (i) A base is a compound of oxide or hydroxide which react with an acid to form salt and water only while an alkali is a soluble base 01 Mark

(ii) An atom is a smallest particle of an element with all chemical properties of an element While Isotopes are atoms of the same element with different number of neutrons (different mass) but have the same atomic number.

01 Mark

(b) Data.

% Composition of C = 52.2%

% Composition of H = 13%

% Composition of O = 34.18%

Vapour density = 23

Empirical formula

Element	C	H	O	
% Composition	52.2	13	34.18	00½
R.A.M	12	1	16	00½
Divide by R.A.M	$\frac{52.2}{12}=4.35$	$\frac{13}{1}=13$	$\frac{34.18}{16}=2.18$	00½
Divide by smallest number	$\frac{4.35}{2.18}$	$\frac{13}{2.18}$	$\frac{2.18}{2.18}$	00½
Ratio into whole number	2	6	1	00½

Empirical formula C_2H_6O 01 Mark

But,

Molecular formula = n(Empirical formula)

Molar mass = 2 × Vapour density. 00½ Mark

$n(C_2H_6O) = 2 \times 23$

$n(12\times2 + 1\times6 + 16) = 46$

$\frac{46n}{46} = \frac{46}{46}$

$n = 1$. 00½ Mark

From

Molecular form = n(Empirical formula)

= 1 (C_2H_6O)

∴ Molecular formula is C_2H_6O. 01 Mark

(c) Oxidation number of nitrogen in KNO_3

Let

$KNO_3 = 0$

$1 + N + (-2 \times 3) = 0$ 00 $\frac{1}{2}$ Mark

$1 + N - 6 = 0$

$N = 6 - 1$

$N = +5$

∴ Oxidation number of N in KNO_3 is +5. 00 $\frac{1}{2}$ Mark.

4. (i) Balanced equation

$$2HCl_{(aq)} + X_2CO_{3(aq)} \rightarrow 2XCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}$$

01 Mark

(ii) Data.

Molarity of acid $(M_a) = 0.1\,M$

Volume of acid $(V_a) = 50\,cm^3$

Number of moles of base $(N_b) = 01$ 00 $\frac{1}{2}$ Mark

Number of moles of acid $(N_a) = 2$

Volume of base $(V_b) = 25$

Molarity of base $(M_b) = ?$

From

$$\frac{M_a V_a}{N_a} = \frac{M_b V_b}{N_b}$$ 00 $\frac{1}{2}$ Mark

Molarity of base $(M_b) = \frac{M_b V_b N_a}{V_a N_b}$

$= \left(\frac{0.1 \times 25 \times 2}{50 \times 1}\right) M$

00 $\frac{1}{2}$ Mark

$= 0.1\,M.$

∴ Concentration of $X_2CO_3 = 0.1\ mol/dm^3$. 01 Mark.

(iii) Relative molecule mass of X_2CO_3

Concentration $= \frac{\text{Mass}}{\text{Volume}}$ 00½ Mark

$= \frac{5.3\ g}{0.5\ L}$.

$= 10.6\ g/dm^3$. 01 Mark

from

Molarity $= \frac{\text{Conc.}\ (g/dm^3)}{\text{Molar Mass}}$ 00½ Mark

Molar Mass $= \left(\frac{10.6}{0.1}\right) g/mol$

∴ Molar Mass of $X_2CO_3 = 106\ g/mol$. 01 Mark

(iv) The name of element X

$X_2CO_3 = 106$ 00½ Mark

$2X + 12 + 3\times16 = 106$

$2X = 106 - 60$

$\frac{2X}{2} = \frac{46}{2}$ 01 Mark

$X = 23$

∴ Element X is sodium since its R.A.M is 23 01 Mark.

5. (a) Three ways of increasing the production of methanol
(i) By increasing pressure
(ii) By lowering the temperature. 1a@ = 03 Marks
(iii) By sucking out Methanol from the system.
(iv) Increasing concentration of carbonmonoxide and hydrogen

(b)(i) Equilibrium position will shift from ~~Right~~ Left handside to ~~left~~ right hand side (~~Backward~~ Forward reaction will be favoured)

(ii) The equilibrium position will shift from Right handside to the ~~Left~~ hand side (Backward reaction will be favoured)

(iii) The equilibrium will shift from Left handside to Right hand side.
1@ = 03 Mark.

(c)(i) The system should be enclosed.
(ii) The reaction should be reversible.
(iii) Factors affecting equilibrium should be kept constant.
(iv) The reaction should be dynamic 1@ = 03 Marks

6. (a)(i) Fertilizers are inorganic substances that contain one or more plant nutrients in concentrated form.
01 Mark

(ii) Three ~~effects~~ of excessive nitrogen to plants
- Delay plant Maturity.

(Pg 5)

- Weakens the plants leading to lower yield and quality
- Reduces sugar contents in plant.

00 1/2 @ = 01 1/2 Mark

b(i) - Banding method
- Broadcasting method
- Spraying method / Follial application.
- Placement
- Drip application

@ 1/2 = 02 (Four points)

(ii) Organic manure are richer in plant nutrients as compared to synthetic manure/fertilizer.
- Improves soil structure
- Organic manure improves water holding capacity by forming humus.
- Organic manure helps in soil erosion control by reducing surface water run off.
- Are not readily reached easily
- Organic manure can stay in the soil for more than one growing season.
- Organic manure attracts microbial activities.
- Do not change the pH of the soil

00 1/2 @ = 02 Marks.

(iii) % Composition by mass = Mass of N in the cpd.

% Composition by Mass = $\frac{\text{Mass of N in the cpd.}}{\text{Mass of compound}} \times 100\%$

00 1/2 Mark.

$M(NH_4)_2SO_4 = (14\times1) + (1\times4)2 + (32\times1) + (16\times4)$

$= (14\times4)2 + 32 + 64$

$= 36 + 32 + 64$

$= 132\ g/mol.$

01 Mark

Mass of N in the compound $14 \times 2 = 28 g/mol$

The percentage composition by mass of N $= \frac{28}{132} \times 100\%$

$= 21.21\%$ 01 Mark

Hence percentage composition of N is 21.21%

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7 (a) (i) Galvanization / Painting
(ii) Anodizing or sacrificial protection
(iii) Painting or sacrificial protection 01 @ = 04 Marks
(iv) Oiling / Greasing

(b) Types of fire extinguishers.
(i) Foam type fire extinguisher
(ii) Carbondioxide fire extinguisher
(iii) Dry chemical
(iv) Wet chemical
(v) ABC extinguisher. 01 @ = 05 Mark
(vi) water AFFW
(vii) sand (viii) - Fire blanket

8. (a) (i) A - $CuCO_3$.
B - CO_2
C - CuO 00½ @ = 02
D - $CuSO_4$

(ii) $CuCO_{3(s)} \xrightarrow{\Delta} CuO_{(s)} + CO_{2(g)}$ 01 Mark.

(b) (i) The sugar change into brown and finally black.
It removes water from sugar and leave carbon as black Mass. 02 Mark

ie $C_{12}H_{22}O_{11(s)} \xrightarrow{\text{Conc. } H_2SO_4} 12C_{(s)} + 11H_2O_{(l)}$

(ii) When conc. sulphuric acid is added to hydrated copper(II) sulphate it change from blue to white. ~~Sulphuric acid~~ 02 Mark

$CuSO_4 \cdot 5H_2O \xrightarrow{\text{conc. } H_2SO_4} CuSO_4 + 5H_2O$.
(Blue) white

(iii) When concentrated sulphuric acid is added to Sulphur, it oxidizes Sulphur to sulphur dioxide.

$S_{(s)} + H_2SO_4{}_{(l)} \longrightarrow 3SO_{2(g)} + 2H_2O_{(l)}$.

02 Mark

SECTION C.

9. (i) Treating insect sting
(ii) Relieving indigestion
(iii) Management of Soil pH
(iv) Reducing acidic rain
(v) Treating industrial wastes
(vi) Neutralize anti acid pills.
vii) Manufacture of fertilizer

3 @ = 15
(Any five points with brief explanation)

10. - Cause soil erosion
- Loss of habitat to some organism, acidic rain, reduction of area of settlement.
- It results into lung diseases due to prolonged exposure to coal dust in mines.
- It cause global warming.
- Air pollution
- Causes water pollution

(3 @ = 15 With explanation)

11. (a)(i) Oxidation half reaction ($CuSO_4$) (Anode)

$4OH^- \longrightarrow 2H_2O + O_2 + 4e^-$ 01 Mark

Reduction half reaction (Cathode)

$Cu^{2+} + 2e^- \longrightarrow Cu(s)$ 01 Mark

General reaction

$2Cu^{2+}_{(aq)} + 4OH^-_{(aq)} \longrightarrow 2Cu_{(s)} + 2H_2O_{(l)} + O_{2(g)}$. 02 Marks.

(ii) Data

Volume of hydrogen = $8.4 dm^3$

GMV volume = $22.4 dm^3$

Coulombs = 96500C 01 Mark

from

$4OH^-_{(aq)} \longrightarrow H_2O_{(l)} + O_{2(g)} + 4e^-$ 01 Mark

$1F = 1e = 96500\ C$

$4F \equiv 44.8 dm^3$ of H_2.

$x \equiv 8.4 dm^3$ of H_2 01 Mark

$x = \left(\frac{4 \times 8.4}{44.8}\right) F.$

$= 0.75F$

The quantity of electricity used is 0.75F 01 Mark.

Pg 9.

If 4F produced $22.4\,dm^3$ of O_2

$0.75F \equiv x$ 01 Mark.

$$x = \left(\frac{0.75 \times 22.4}{4}\right) dm^3$$

$$= 4.2\,dm^3.$$

∴ The volume of oxygen that will be produced is $4.2\,dm^3$. 01 Mark.

(b) The mass of copper produced

Eq. $Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$

$2F \equiv 63.5g$ of Cu 01 Mark

$0.75F \equiv x$

$$x = \left(\frac{0.75F \times 63.5g}{2F}\right)$$

$$= 23.81\,g.$$

∴ The mass of copper produced was 23.81 g. 01 Mark

Pg 10.